3D Dynamic Reconstruction from Multi-view Capture

Sinuo Liu

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Abstract

This thesis addresses the poor reconstruction quality of mesh-bound Gaussian methods in regions undergoing topology changes, proposes a scene-flow discontinuity measure to detect such regions and adapt the optimization accordingly, and evaluates the approach against GauSTAR on multi-view human sequences.

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Chapter 1

Introduction

Identify the limitations of GauSTAR, namely its reliance on depth for topology detection and its uniform treatment of all faces, and list the contributions of this thesis.

Chapter 2

Related Work

2.1 Dynamic Reconstruction with Gaussian Splatting

Review 3D Gaussian Splatting and mesh-bound Gaussian methods for dynamic scenes, with a focus on GauSTAR.

2.2 Optical Flow and Scene Flow

Review optical flow and point tracking methods, including RAFT, SEA-RAFT, CoTracker3 and CoWTracker.

2.3 Monocular Depth Estimation

Briefly introduce monocular depth models such as Depth Anything 3 as a potential replacement for sensor depth.

Chapter 3

Preliminaries

3.1 Pipeline

Describe the GauSTAR pipeline, including the fixed-topology stage, Gaussian unbinding and topology update.

3.2 Losses and Topology Detection

Introduce the normal consistency and depth losses, as well as the original depth-based topology error detection.

Chapter 4

Method

4.1 Flow Backbone Selection

Compare flow methods by end-point error against ground-truth flow derived from tracked meshes, and justify the choice of RAFT / SEA-RAFT.

4.2 Scene Flow Discontinuity for Topology Detection

Localize topology-changing regions by measuring the inconsistency of 3D motion between neighboring faces, including k-ring propagation and normalization.

4.3 Topology-Adaptive Optimization

Increase the normal consistency weight in detected topology-changing regions and drive the Gaussian unbinding weights with the scene flow discontinuity score.

4.4 Depth-Free Variant

Remove the sensor depth loss and replace sensor depth with Depth Anything 3 predictions calibrated by a bilateral grid.

Chapter 5

Experiments

5.1 Setup

Describe the multi-view dataset (Takes 3, 4, 8 and 12), the evaluation metrics (PSNR, Chamfer distance, etc.) and implementation details.

5.2 Comparison with GauSTAR

Compare the proposed method with the original GauSTAR quantitatively and qualitatively on all sequences.

5.3 Ablation Studies

Analyze the effect of the discontinuity measure, weighting strategy, unbinding formulation and depth source.

5.4 Limitations

Present failure cases, such as artifacts at the inner elbow, and discuss the current limitations of the method.

Chapter 6

Conclusion

Summarize the main findings and outline directions for future work.